

National University of Computer and Emerging Sciences
Karachi Campus

Computer Networks (CS3001)

Final Exam

Date: May 19th 2025

Course Instructor

Dr. Ghufraan Ahmed, Dr. Fahad Samad,
Dr. Aqsa, Dr. Farrukh, Ms. Hira, Ms. Yusra,
Ms. Rabia, Mr. Shoaib Raza, Mr. Usama,
Mr. Usman, Mr. Waseem, and Mr. Sandesh

Total Time (Hrs): 03

Total Marks: 50

Total Questions: 05

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions in the given sequence.

CLO # 2: Demonstrate the basics of network concepts using state-of-the-art network tools.

Question # 1:

[40 Minutes]

[20 x 0.5 = 10 Marks]

Give your answer as true or false to show your agreement or disapproval to the following along with one sentence justification. Note: No points will be awarded without justification.

- i. Packet switching is more efficient than circuit switching for bursty data traffic.
- ii. The transport layer provides services directly to the user.
- iii. The Domain Name System (DNS) uses a centralized database to store IP-address mappings.
- iv. In peer-to-peer architecture, each peer can act as both a client and server.
- v. In TCP's slow start phase, the congestion window increases linearly.
- vi. The control plane determines how datagrams are forwarded at each router.
- vii. Middleboxes include devices such as firewalls, NATs, and load balancers.
- viii. DASH (Dynamic Adaptive Streaming over HTTP) adjusts video quality based on network conditions.
- ix. OSPF is used for routing between autonomous systems.
- x. Software-defined networking (SDN) centralizes the control plane.
- xi. MAC addresses are used to identify devices across the Internet.
- xii. Cookies cannot be used to track user behavior across multiple websites.
- xiii. The HEAD method in HTTP is used to retrieve only metadata, not the body of a resource.
- xiv. A CNAME record in DNS maps a domain name to an IP address.
- xv. The Border Gateway Protocol (BGP) uses distance-vector routing within an autonomous system.
- xvi. The Border Gateway Protocol (BGP) guarantees loop-free paths using hop counts.
- xvii. A routing algorithm that always chooses the least congested path is OSPF.
- xviii. A MAC address is globally unique and remains unchanged regardless of location.
- xix. The Ethernet protocol uses a connection-oriented communication model.
- xx. A switch creates a broadcast storm by default.

National University of Computer and Emerging Sciences

Karachi Campus

CLO # 2: Demonstrate the basics of network concepts using state-of-the-art network tools.

Question # 2:

[40 Minutes]

[5 x 2 = 10 Marks]

Part (a) The FAST-NUCES Karachi campus is experiencing intermittent network slowdowns during peak hours, especially between 10:00 AM and 2:00 PM. As the newly hired Network Engineer, you analyze the traffic patterns and notice that one of the core router is transmitting packets, each of length 8000B, over a single link with transmission rate 100 Mbps to another router at the other end of the link with an average packet arrival rate traffic is 30packets/sec. What is the Traffic Intensity and does it suggest high queuing delay?

Part (b) Susan browses nytimes.com, which contains an ad loaded from adx.com. As a result, her browser receives a third-party cookie from AdX [Set-Cookie: ID=7493; Domain=adx.com]. This cookie is stored in her browser under the domain adx.com. The next day, Susan visits socks.com, which also loads ads from AdX. Her browser sends the ID=7493 cookie back to adx.com automatically. Using this cookie, AdX recognizes Susan and serves her a personalized ad for running socks based on her past browsing on nytimes.com.

- i. What kind of cookie is being used by adx.com?
- ii. Why is adx.com able to identify Susan when she visits socks.com, a completely different website?

Part (c) Bob clicks a video link on netcinema.com with the URL: <http://netcinema.com/6Y7B23V>

The following steps occur:

Bob's browser contacts his local DNS server to resolve the domain.

The local DNS contacts netcinema.com's authoritative DNS.

Instead of an IP address, the DNS response includes a CNAME record pointing to: <http://KingCDN.com/NetC6y&B23V>

Bob's browser resolves KingCDN.com via a second DNS query.

After obtaining the IP of the CDN edge server, the video is requested and streamed via HTTP.

- i. Why does netcinema.com return a CNAME pointing to KingCDN.com instead of a direct IP? From which entity does Bob ultimately download the video file?
- ii. What is the advantage of this two-step DNS redirection process for netcinema.com?

Part (d) A startup called StudyZone has registered the domain studyzone.org through BlueHost, which also hosts their authoritative name server.

The web server for StudyZone.org is located at IP address: 192.168.100.20

The mail server for StudyZone is hosted at: 192.168.100.30

The authoritative name server is dns1.bluehost.com, and it maps to IP address: 203.0.113.10

List down all the resource records (RRs) that will be inserted into the authoritative name server and .com's TLD (top level domain) server.

Part (e) A satellite-based communication company provides data transmission services using a reliable data transfer protocol (RDT 3.0), which employs a stop-and-wait mechanism for reliability. Due to the satellite link, the average round-trip time (RTT) is high measured at 250 milliseconds. The company sends data packets of size 10 KB over a link with a bandwidth of 1 Mbps. Calculate the channel utilization and then explain the result.

National University of Computer and Emerging Sciences

Karachi Campus

CLO # 4: Apply Socket Programming (client/server) to solve various real-world problems, including ensuring of data integrity.

Question # 3:

[70 Minutes]

[5 x 4 = 20 Marks]

Part (a) Below is the pseudo code that describes the reliable data transfer part of the TCP sender. The mechanism to adjust the window size for congestion control is missing. Please identify the code block where the missing functionalities shall be added and justify your pick.

```

NextSeqNum = InitialSeqNum
SendBase = InitialSeqNum
CongestionWindowSize = InitialCongestionWindowSize

loop (forever) {
    switch(event)
    event: data received from application above
        create TCP segment with sequence number NextSeqNum
        If (NextSeqNum < SendBase + CongestionWindowSize) {
            if (timer currently not running)
                start timer
            pass segment to IP
            NextSeqNum = NextSeqNum + length(data)
        } else
            refuse data
    event: timer timeout
        retransmit from the smallest, not-yet-acknowledged segment
        start timer
    event: ACK received, with ACK field value of y
        if (y > SendBase) {
            SendBase = y
            if (there are currently not-yet-acknowledged segments)
                (re)start timer
        } else {
            increment count of dup ACKs received for y
            if (count of dup ACKs received for y = 3) {
                resend segment with sequence number y
            }
        }
} /* end of loop forever */

```

A {

B {

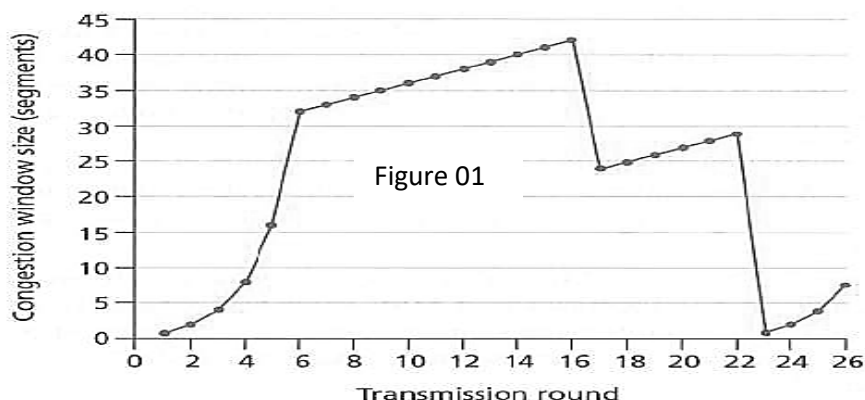
C {

D {

E {

- Where to add the code to increase the congestion window size? Why?
- Where to add the code to reduce the congestion window size to 1? Why?
- Where to add the code to reduce the window size to half? Why?
- Where to add the code to set the slow start threshold to half of the window size? Why?

Part (b) Assume that TCP at a host is behaving as shown in Figure 01 Answer the following questions:



National University of Computer and Emerging Sciences

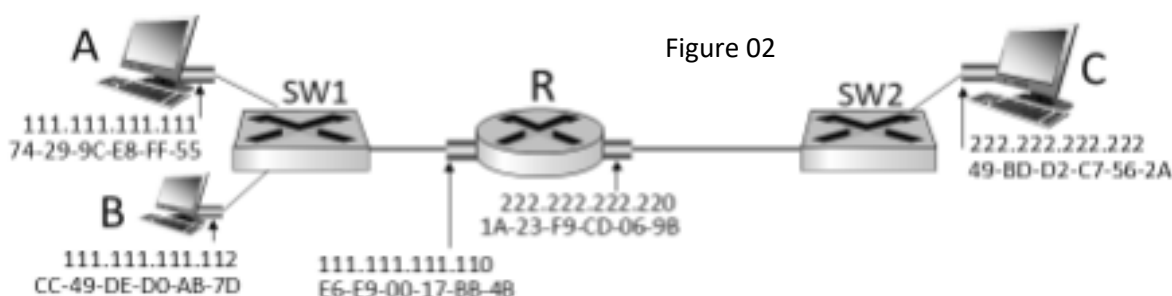
Karachi Campus

- i. List various TCP phases with transmission round ranges.
- ii. Explain the transitions in round 6, 17 and 23.
- iii. In which round, the 66th TCP segment will be transmitted.
- iv. Find the congestion window size at the end of the 10th transmission.

Part (c) NUCES Karachi campus, School of Computing has been assigned the network 208.16.7.0/24 and wants to divide it among the five Departments. Computer Science (CS) needs to accommodate up to 122 hosts, Software Engineering (SE) needs to accommodate up to 58 hosts, Artificial Intelligence (AI) needs to accommodate up to 30 hosts, and Cyber Security (CY) needs to accommodate up to 12 hosts. Fill the following table 01 in your answer script with the details of the sub-networks that the NUCES can create to fit its Department's needs.

Departments	Network Address	Net Mask	Usable IP range	Broadcast Address
CS				
SE				
AI				
CY				

Part (d) Consider Figure 02 and answer the following questions:



- i. Consider sending an IP datagram from PC-A to PC-B. PC-A issues an ARP request because it needs to send a packet to PC-B. In this scenario, what will happen next?
- ii. Suppose PC-A would like to send IP datagram to PC-C on a remote LAN, however there are currently no mappings in the ARP cache. How will the device obtain a destination MAC address?
- iii. What will be the content of the Source and destination IP-address field in the header of the IP datagram sent by PC-A to PC-C?
- iv. What will be the content of the Source and destination MAC-address field in the header of the IP datagram sent by PC-A to PC-C?

CLO # 3: Demonstrate operations of various classical routing and switching protocol via simulators.

Question # 4:

[30 Minutes]

[4 + 2 + 4 = 10 Marks]

Part (a) For the network graph given in Figure 03, you are required to identify and draw Least-cost-path tree from node Y and resulting forwarding table in Y. Do show the process through each iteration using table 02.

Step	Node (N')	D(u), p(u)	D(v), p(v)	D(w), p(w)	D(x), p(x)	D(z), p(z)
0	y	∞	∞	∞	∞	∞

National University of Computer and Emerging Sciences

Karachi Campus

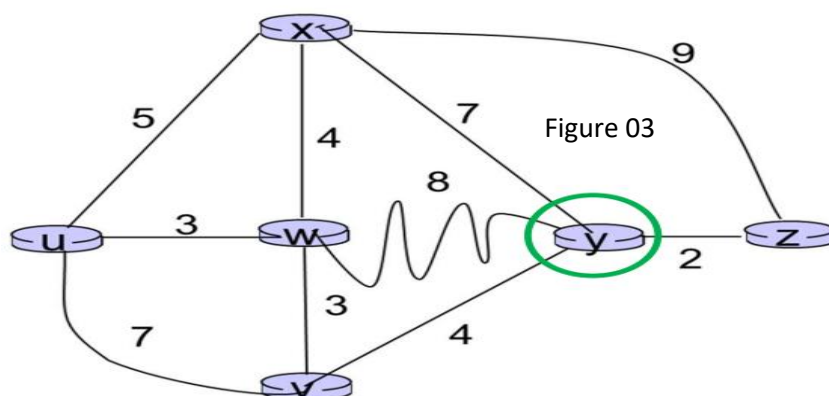


Figure 03

Part (b) Now apply Distance Vector Routing Algorithm on the network graph given in Figure 03 to answer the following questions.

- i. What are the initial distance vectors for router 'w'? Write your answer as u, v, w, x, y, and z and if a distance is ∞ , write ∞ .
- ii. When the algorithm converges, what are the distance vectors from router 'y' to all routers? Write your answer as u, v, w, x, y, and z.

Part (c) For the network shown in Figure 04, suppose that the desired forwarding behavior is that packets from Host H1 or Host H2 destined to Host H5 or Host H6 are to be forwarded from s3 to s1, and then from s1 to s2 (thus completely avoiding the use of the link between s3 and s2).

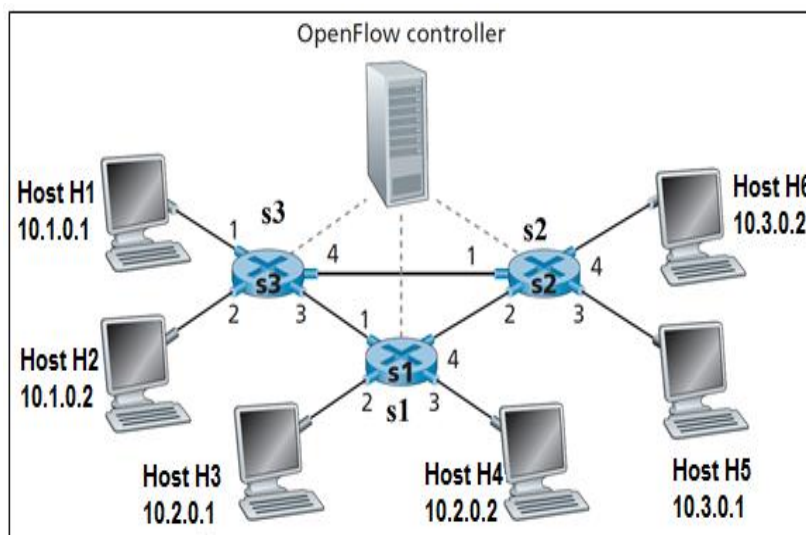


Figure 04

- i. How the SDN data plane is different from SDN control plane? Explain.
- ii. Write down the flow table entry for s3, so that datagram sent from Host H1 or Host H2 are forwarded to s1 over interface 3.
- iii. Write down the flow table entry for s1, so that datagram arriving at port 1 of s1, from s3, are forwarded to s2 over outgoing interface 4.
- iv. Finally write down the flow table entry for s2, for forwarding to required destinations.

BEST OF LUCK ☺